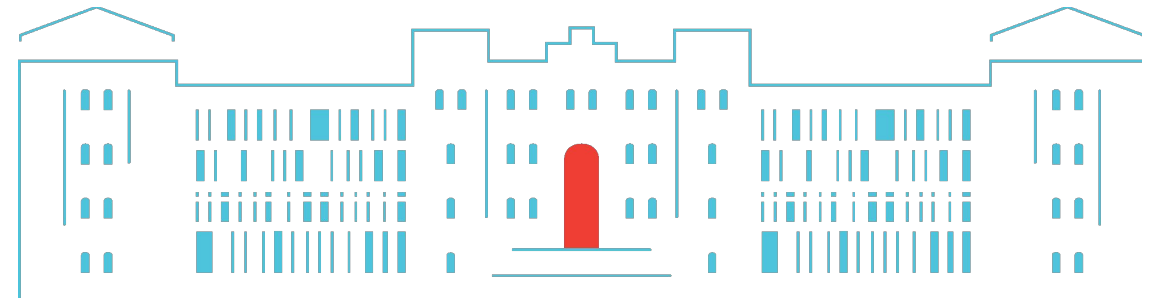


Localization of mobile robots coupling building information modeling and Wi-Fi-based positioning

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Technische
Universität
Hamburg

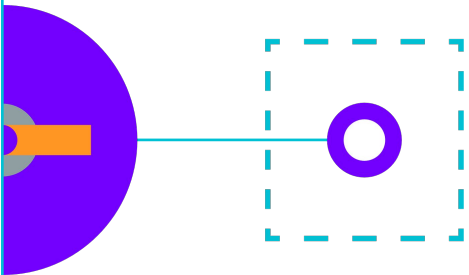
10.03.2026



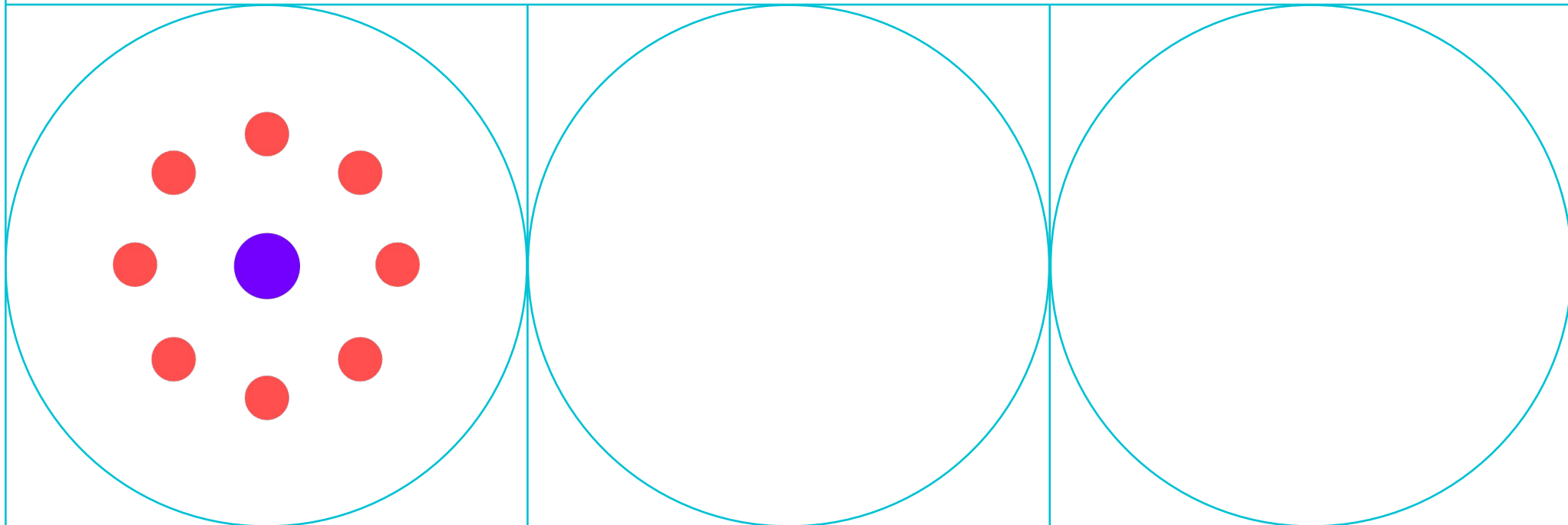
Kavya Dave - 565772

Agenda:

1. Motivation and background
2. Research problem and goal
3. Research gap
4. Proposed hybrid framework pipeline
5. Implementing the framework
6. Validation and results
7. Conclusion and future works
8. References



1. Motivation and background



Motivation and background

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- The rise of industry 4.0 :- mobile robots for automated inspections and monitoring task
- Requirement :- reliable, continuous, and precise positioning
- Global navigation satellite system (GNSS) failure :- GNSS works for outdoors, but concrete and metal block the signals

2. Research problem and goal



Research problem and goal

Research Problem

- Indoors suffer from signal attenuation (weakening), multipath propagation (reflections), and NLOS conditions (blocked paths)

Goal

- centimeter-level accuracy

Competitive Landscape: Technology Comparison

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LiDAR/Vision

- ✓ High precision
- ⚠ Computationally heavy
- ⚠ Lighting dependent



UWB

- ✓ High accuracy (10-30 cm)
- ⚠ High hardware costs
- ⚠ Limited range



BLE/ RFID

- ✓ Low cost/ power
- ⚠ Restricted by unstable RSSI
- ⚠ Lower accuracy (1-5 m)

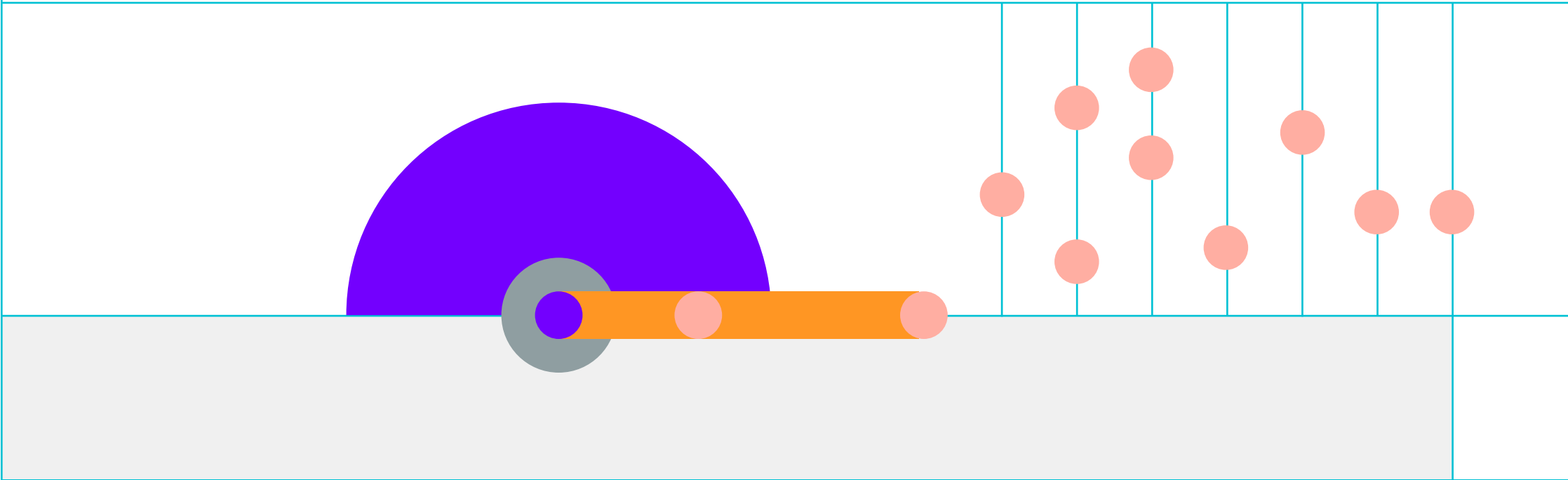


Why Wi-Fi

- ★ **Ubiquity:** Leverages existing, already deployed building infrastructure.



3. Research gap

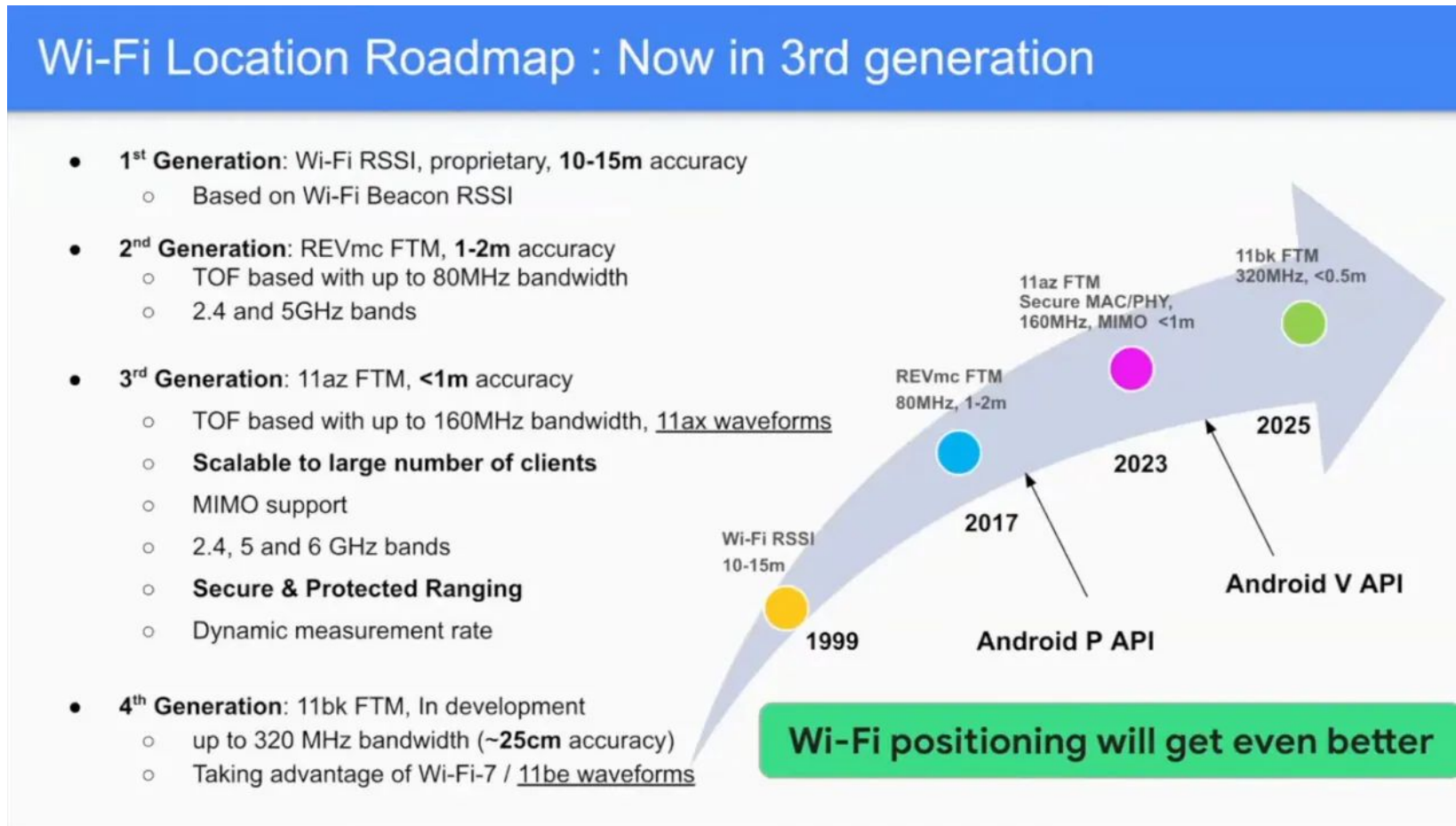


Research gap

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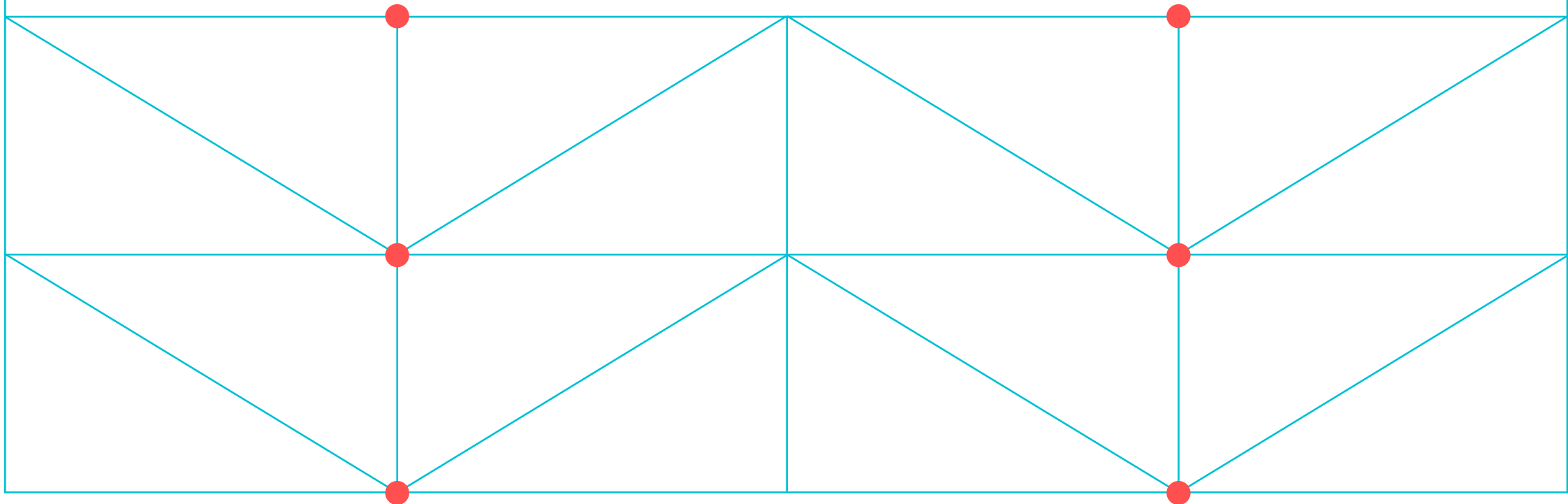
- Fragmented data :- received signal strength indicator(RSSI), which is too erratic
- Blind infrastructure :- no optimization
- Environmental ignorance :- ignore walls or materials

Evolution of Wi-Fi



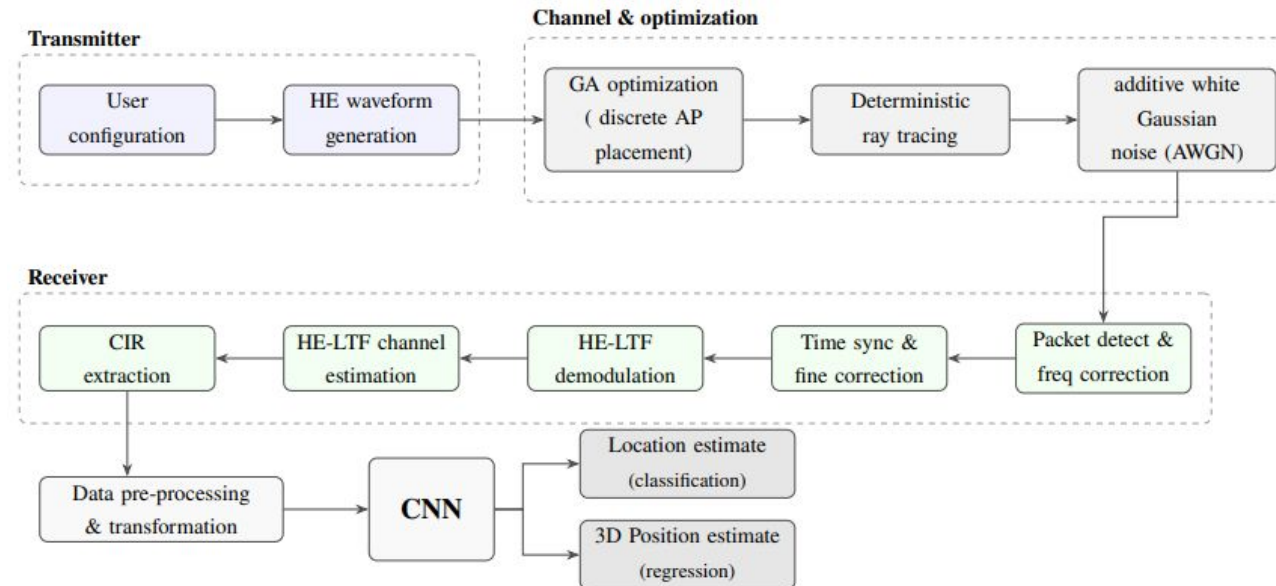
Wi-Fi roadmap – taken from Ref [1]

4. Proposed hybrid framework pipeline



Proposed hybrid framework pipeline

- 4- stage design :
 - 1) optimization : planning the AP layout
 - 2) Simulation : creating the digital twin environment
 - 3) Fusion : combining the signal characteristics
 - 4) Dual-head output : which zone or room, precise coordinate



End-to-end pipeline

AP deployment via discrete optimization

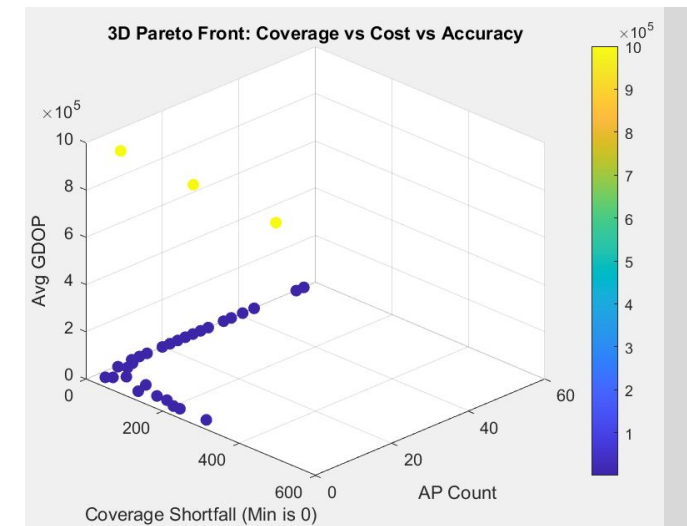
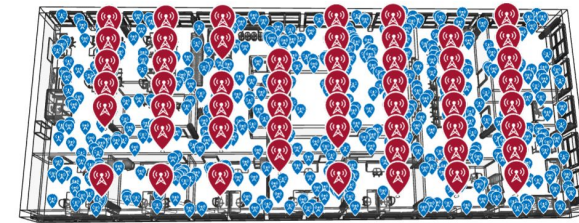
Discrete optimization strategy (DOS)

Problem : placing APs randomly leads to blind spots behind the walls

Brain of Planning : MOGA

- optimize for maximum coverage, minimum cost(fewer APs), and best geometry (geometric dilution of precision)(GDOP)
- **Knee-point selection** picking the sweet spot on the graph where we get the most accuracy for the least money

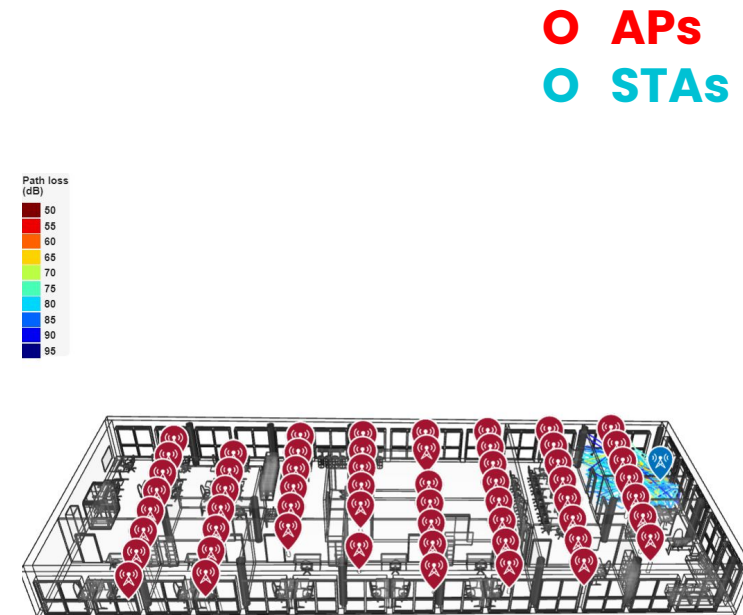
DOS with 0.5m separation



Knee-point, where deep blue represents excellent geometry (low GDOP), while bright yellow represents poor geometry (high GDOP)

Building information modeling (BIM) -assisted ray tracing

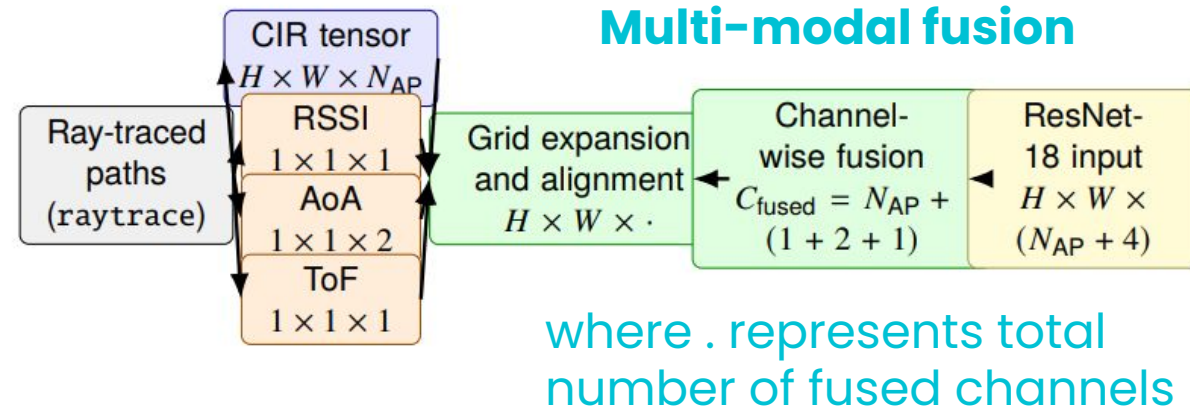
- Digital twin : we use STL files from BIM to model concrete, metal, glass, wood and plasterboard
- 5 bounces using the real building materials



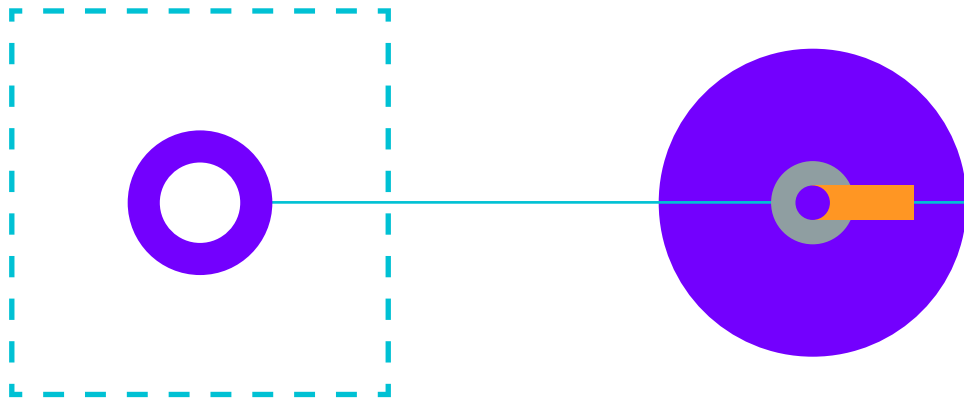
Ray tracing

Multi-modal feature fusion

- Step 1 : ray trace
- Step 2 : feature extraction
- Step 3: grid expansion
- Step 4: channel-wise fusion
 - Concatenates channel impulse response (CIR), RSSI, angle of arrival (AoA), and time of flight (ToF) into a single 4D tensor
- Step 5 : Resultant tensor size for residual network (ResNet-18) as input : $H \times W \times (N_{AP} + 4)$



5. Implementing the framework

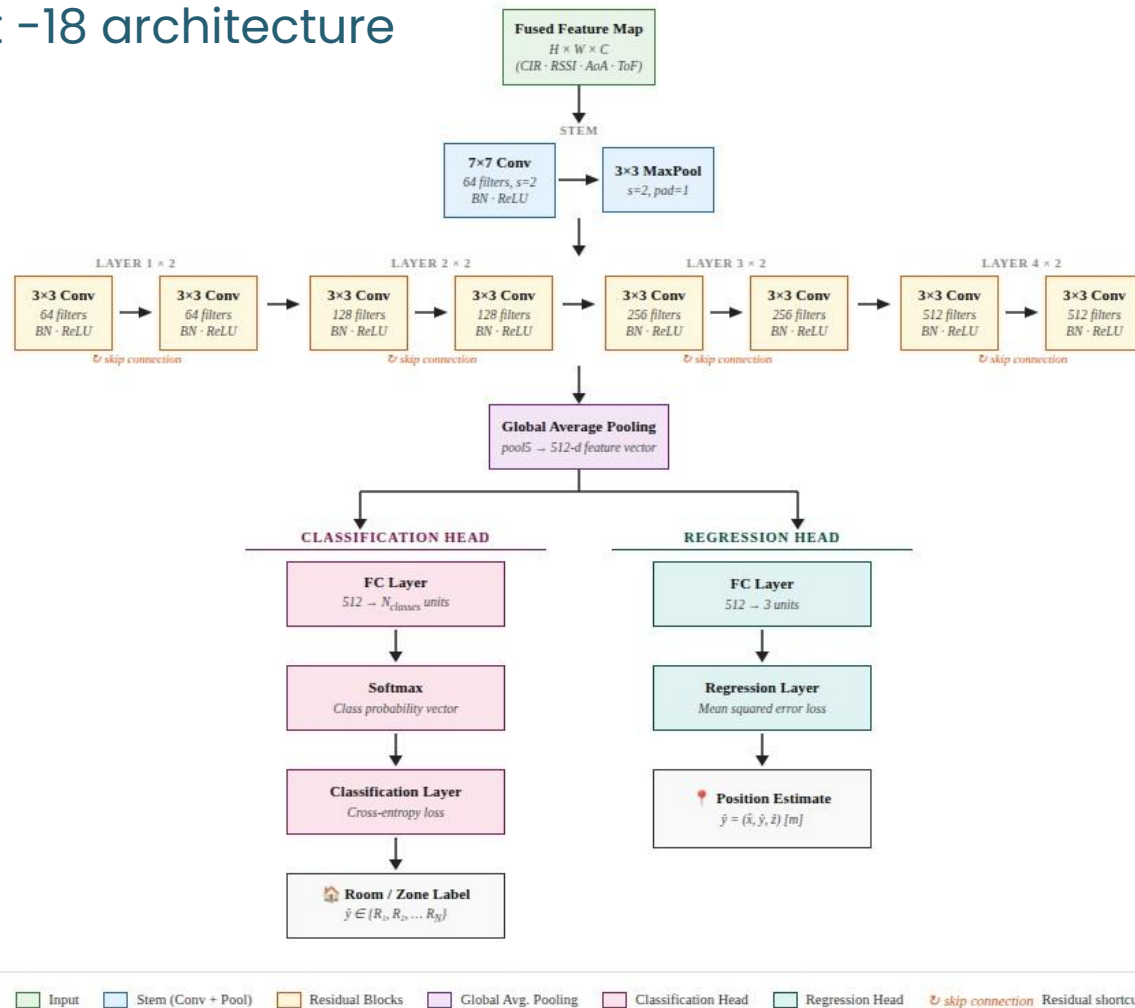


Implementing the framework

- Implementing the framework (software / hardware)
 - Managed in MATLAB 2023b using WLAN and deep learning toolboxes
- Data generation and SNR diversity
 - simulate real-world noise
 - Data augmentation

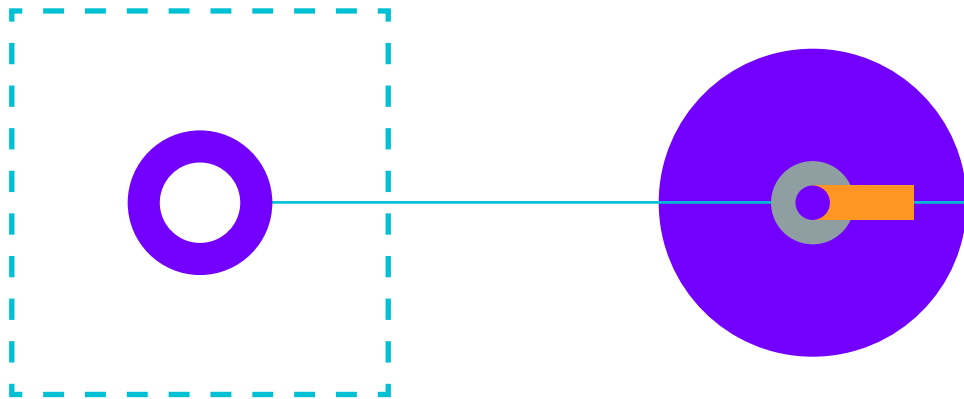
ResNet-18

ResNet -18 architecture



- Combined loss function
 - $$L_{\text{total}} = L_{\text{class}} + (\text{Lambda}) L_{\text{reg}}$$
 - Stabilizes training to provide local coordinate precision and global context

6. Validation and results



Validation and results

- Validation study groups
 - Group 1 : Antenna scaling (4 x 1, 2 x 1, 1 x 1)
 - Group 2 : STA separation (0.5 m vs 0.7 m)
 - Group 3 : Bandwidth (40, 80 and 160 MHz)

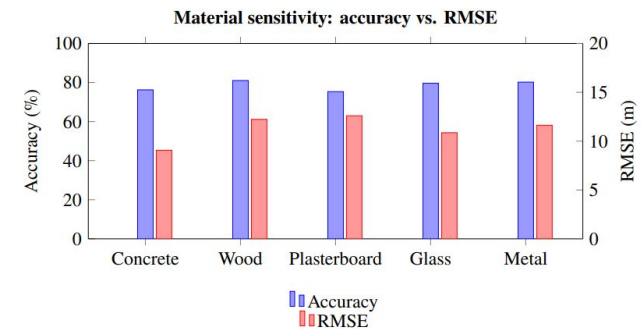
1	0.7	CBW160	4x1	Plasterboard	Random	39 × 15
2	0.7	CBW160	2x1	Plasterboard	Random	39 × 15
3	0.7	CBW160	1x1	Plasterboard	Random	39 × 15
4	0.50	CBW40	4x1	Concrete	Uniform	39 × 15
5	0.70	CBW40	4x1	Concrete	Uniform	39 × 15
6	0.5	CBW80	4x1	Metal	Uniform	39 × 15
7	0.5	CBW160	4x1	Metal	Uniform	39 × 15
8	0.5	CBW40	4x1	Metal	Uniform	39 × 15

Dataset configuration parameters for indoor positioning

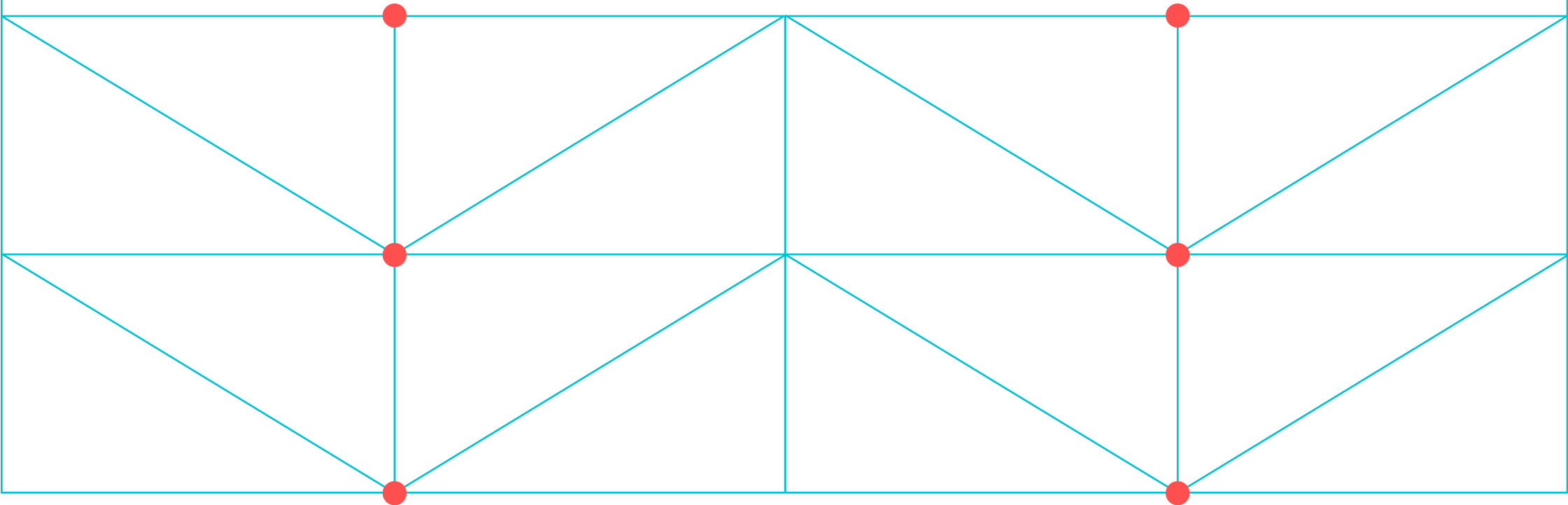
Final evaluation of study groups

Study group	Variable	ID	Accuracy (%)		RMSE (m)	
			Baseline	Proposed	Baseline	Proposed
1. Antenna scaling	1×1	3	55.07	83.06	9.19	7.95
	2×1	2	57.58	82.52	7.33	4.43
	4×1	1	42.16	76.83	10.21	7.90
2. STA separation	0.50 m	4	35.68	67.27	17.83	15.89
	0.70 m	5	36.24	69.87	39.895	33.93
3. Bandwidth	CBW40	6	38.31	65.70	15.45	14.87
	CBW80	7	34.80	59.22	19.39	16.12
	CBW160	8	35.52	65.01	11.22	10.56

Where baseline is CIR_only and proposed is CIR_ToF_AoA_RSSI



7. Conclusion and future work



Conclusion and future work

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- Optimal configuration is 2 x 1 with 0.5 m separation distance and 160MHz
- Discrete optimization strategy combined with proposed multi-modal fusion offers a solid foundation
- However, the proposed framework is unable to achieve centimeter-level accuracy mainly because of signal reflections, bandwidth limits and grid constraint
- Future : Integration of inertial/ IMU data to fix drift and testing on commercial 802.11az hardware

References

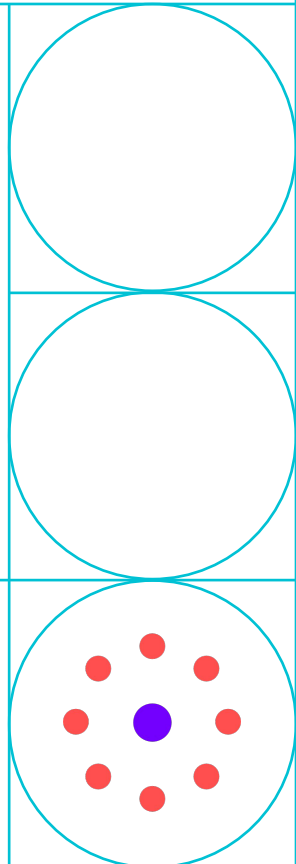
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[1] Rahman , M. (2024, November 12). Android 15 adds support for Wi-Fi ranging to improve indoor navigation. Android authority.
<https://www.androidauthority.com/android-15-wi-fi-ranging-3498218/>

Dankeschön

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